Unit Test Report

Project: Customer Churn Prediction (Team Hackforge)
Course: Software Engineering Mini Project, Deliverable Part 3
Date of test run: 30 October 2026
Generated by: `python tools/generate_test_reports.py` from a single run of the automated test suite

1. Purpose

This report records the result of every automated test in the project. The tests are written with pytest and cover the prediction pipeline, the Streamlit user interface, the training pipeline, and the deep learning and quantum models. Each test is linked to the Test Plan v1.1 test case it executes.

2. Test Environment

Item	Value
Operating system	Darwin 25.3.0 (arm64)
Python	3.12.7
pandas	3.0.6
numpy	2.5.4
scikit-learn	1.9.1
xgboost	3.4.1
streamlit	1.65.0
mlflow	3.17.2
dvc	3.67.1
torch	2.15.0
pennylane	0.45.1
pytest	9.1.1
pytest-cov	7.1.0
pylint	4.1.3

Command: `pytest --cov --cov-branch --junitxml=docs/reports/test-reports/raw/junit.xml`

3. Summary

Total tests	Passed	Failed	Skipped	Pass rate	Duration
152	152	0	0	100.0%	71.5 s

4. Results by Test Module

Module	What it tests	Tests	Passed	Failed	Skipped	Time
`test_pipeline.py`	Prediction pipeline: artefact loading, validation, preprocessing, prediction, metrics	84	84	0	0	1.4 s

Module	What it tests	Tests	Passed	Failed	Skipped	Time
`test_app.py`	Streamlit user interface, run headless with AppTest	12	12	0	0	12.0 s
`test_sprint2.py`	Batch scoring, churn driver explanation and what-if simulator	4	4	0	0	0.1 s
`test_paradigms.py`	Deep learning (DML) and quantum (QML) models, paradigm comparison	19	19	0	0	2.8 s
`test_system_validation.py`	End-to-end training pipeline, benchmark and non-functional checks (TC-01 to TC-13)	33	33	0	0	45.1 s

5. Results by Test Level

Level	Tests	Passed	Failed
Integration	27	27	0
Non-functional	13	13	0
System	38	38	0
Unit	74	74	0

6. Detailed Results

No.	Module	Test	Test case	Result	Time (s)
1	test_app	`test_app_starts_with_default_preset`	TC-14	Passed	1.71
2	test_app	`test_every_input_has_label_and_help`	TC-14	Passed	0.51
3	test_app	`test_loaded_preset_is_used_for_prediction`	TC-16	Passed	1.29
4	test_app	`test_form_values_survive_page_switch`	TC-14	Passed	1.04
5	test_app	`test_cleared_form_reports_missing_fields`	TC-15	Passed	0.84
6	test_app	`test_invalid_combination_shows_error_instead_of_prediction`	TC-15	Passed	0.72
7	test_app	`test_estimated_total_charges`	TC-14	Passed	0.77
8	test_app	`test_live_evaluation_matches_benchmark`	TC-17	Passed	1.21
9	test_app	`test_live_evaluation_custom_threshold`	TC-17	Passed	1.00
10	test_app	`test_live_results_survive_page_switch`	TC-17	Passed	1.20
11	test_app	`test_prediction_shows_all_three_paradigms`	TC-20	Passed	0.97
12	test_app	`test_no_paradigm_comparison_for_invalid_input`	TC-20	Passed	0.74
13	test_paradigms	`test_predict_proba_is_a_valid_distribution[fitted_mlp]`	TC-21, TC-22	Passed	2.45
14	test_paradigms	`test_predict_proba_is_a_valid_distribution[fitted_vqc]`	TC-21, TC-22	Passed	0.12
15	test_paradigms	`test_pickle_round_trip_keeps_predictions[fitted_mlp]`	TC-21, TC-22	Passed	0.02

No.	Module	Test	Test case	Result	Time (s)
16	test_paradigms	test_pickle_round_trip_keeps_predictions[fitted_vqc]	TC-21, TC-22	Passed	0.03
17	test_paradigms	test_early_stopping_restores_best_epoch[fitted_mlp]	TC-21, TC-22	Passed	0.00
18	test_paradigms	test_early_stopping_restores_best_epoch[fitted_vqc]	TC-21, TC-22	Passed	0.00
19	test_paradigms	test_clone_keeps_hyperparameters	TC-21	Passed	0.00
20	test_paradigms	test_mlp_parameter_count_matches_architecture	TC-21	Passed	0.00
21	test_paradigms	test_vqc_parameter_split	TC-22	Passed	0.00
22	test_paradigms	test_vqc_features_are_rotation_angles	TC-22	Passed	0.01
23	test_paradigms	test_single_qubit_circuit_reduces_to_cosine	TC-22	Passed	0.01
24	test_paradigms	test_benchmark_rejects_unknown_paradigm	TC-23	Passed	0.00
25	test_paradigms	test_benchmark_plan_tags_each_paradigm	TC-23	Passed	0.00
26	test_paradigms	test_classical_parameter_count_for_linear_model	TC-23	Passed	0.08
27	test_paradigms	test_parameter_count_of_torch_models	TC-21, TC-22	Passed	0.00
28	test_paradigms	test_saved_paradigm_models_carry_their_test_metrics	TC-20	Passed	0.01
29	test_paradigms	test_predict_all_paradigms_scores_one_customer_with_every_model	TC-20	Passed	0.04
30	test_paradigms	test_threshold_applies_to_every_paradigm	TC-20	Passed	0.04
31	test_paradigms	test_load_paradigm_model_errors	TC-20	Passed	0.01
32	test_pipeline	test_imports	TC-10	Passed	0.00
33	test_pipeline	test_load_artifacts	TC-09	Passed	0.00
34	test_pipeline	test_missing_model_file_handling	TC-09	Passed	0.00
35	test_pipeline	test_missing_encoder_file_handling	TC-09	Passed	0.00
36	test_pipeline	test_valid_customer_input_validation	TC-15	Passed	0.00
37	test_pipeline	test_invalid_numerical_input_negative_tenure	TC-15	Passed	0.00
38	test_pipeline	test_invalid_numerical_input_excessive_charges	TC-15	Passed	0.00
39	test_pipeline	test_invalid_senior_citizen_value	TC-15	Passed	0.00
40	test_pipeline	test_missing_required_fields	TC-15	Passed	0.00
41	test_pipeline	test_feature_ordering_and_preprocessing	TC-16	Passed	0.01
42	test_pipeline	test_unknown_categorical_value_handling	TC-15	Passed	0.01
43	test_pipeline	test_successful_prediction_output	TC-16	Passed	0.01
44	test_pipeline	test_customer_id_is_dropped_if_present	TC-11	Passed	0.01
45	test_pipeline	test_predict_churn_custom_threshold	TC-17	Passed	0.02

No.	Module	Test	Test case	Result	Time (s)
46	test_pipeline	test_threshold_boundary_cases	TC-17	Passed	0.02
47	test_pipeline	test_get_risk_tier_classification	TC-16	Passed	0.00
48	test_pipeline	test_valid_categories_match_fitted_encoders	TC-25	Passed	0.00
49	test_pipeline	test_blank_values_reported_as_missing[None]	TC-15	Passed	0.00
50	test_pipeline	test_blank_values_reported_as_missing[]	TC-15	Passed	0.00
51	test_pipeline	test_blank_values_reported_as_missing[ ]	TC-15	Passed	0.00
52	test_pipeline	test_blank_values_reported_as_missing[nan]	TC-15	Passed	0.00
53	test_pipeline	test_non_dict_input_is_rejected	TC-15	Passed	0.00
54	test_pipeline	test_invalid_category_value_is_rejected	TC-15	Passed	0.00
55	test_pipeline	test_category_with_wrong_type_is_rejected	TC-15	Passed	0.00
56	test_pipeline	test_numeric_fields_reject_bad_types[abc-tenure]	TC-15	Passed	0.00
57	test_pipeline	test_numeric_fields_reject_bad_types[abc-MonthlyCharges]	TC-15	Passed	0.00
58	test_pipeline	test_numeric_fields_reject_bad_types[abc-TotalCharges]	TC-15	Passed	0.00
59	test_pipeline	test_numeric_fields_reject_bad_types[abc-SeniorCitizen]	TC-15	Passed	0.00
60	test_pipeline	test_numeric_fields_reject_bad_types[True-tenure]	TC-15	Passed	0.00
61	test_pipeline	test_numeric_fields_reject_bad_types[True-MonthlyCharges]	TC-15	Passed	0.00
62	test_pipeline	test_numeric_fields_reject_bad_types[True-TotalCharges]	TC-15	Passed	0.00
63	test_pipeline	test_numeric_fields_reject_bad_types[True-SeniorCitizen]	TC-15	Passed	0.00
64	test_pipeline	test_numeric_fields_reject_bad_types[inf-tenure]	TC-15	Passed	0.00
65	test_pipeline	test_numeric_fields_reject_bad_types[inf-MonthlyCharges]	TC-15	Passed	0.00
66	test_pipeline	test_numeric_fields_reject_bad_types[inf-TotalCharges]	TC-15	Passed	0.00
67	test_pipeline	test_numeric_fields_reject_bad_types[inf-SeniorCitizen]	TC-15	Passed	0.00
68	test_pipeline	test_numeric_fields_reject_bad_types[bad_value3-tenure]	TC-15	Passed	0.00
69	test_pipeline	test_numeric_fields_reject_bad_types[bad_value3-MonthlyCharges]	TC-15	Passed	0.00
70	test_pipeline	test_numeric_fields_reject_bad_types[bad_value3-TotalCharges]	TC-15	Passed	0.00
71	test_pipeline	test_numeric_fields_reject_bad_types[bad_value3-SeniorCitizen]	TC-15	Passed	0.00

No.	Module	Test	Test case	Result	Time (s)
72	test_pipeline	test_clean_customer_input_normalises_values	TC-15	Passed	0.00
73	test_pipeline	test_clean_customer_input_keeps_unconvertible_values	TC-15	Passed	0.00
74	test_pipeline	test_numeric_range_boundaries[tenure-0-True]	TC-15	Passed	0.00
75	test_pipeline	test_numeric_range_boundaries[tenure-120-True]	TC-15	Passed	0.00
76	test_pipeline	test_numeric_range_boundaries[tenure-121-False]	TC-15	Passed	0.00
77	test_pipeline	test_numeric_range_boundaries[tenure--1-False]	TC-15	Passed	0.00
78	test_pipeline	test_numeric_range_boundaries[tenure-12.5-False]	TC-15	Passed	0.00
79	test_pipeline	test_numeric_range_boundaries[MonthlyCharges-500.0-True]	TC-15	Passed	0.00
80	test_pipeline	test_numeric_range_boundaries[MonthlyCharges-500.01-False]	TC-15	Passed	0.00
81	test_pipeline	test_numeric_range_boundaries[MonthlyCharges--0.01-False]	TC-15	Passed	0.00
82	test_pipeline	test_numeric_range_boundaries[TotalCharges-50000.01-False]	TC-15	Passed	0.00
83	test_pipeline	test_numeric_range_boundaries[TotalCharges--1-False]	TC-15	Passed	0.00
84	test_pipeline	test_total_charges_inconsistent_with_tenure	TC-15	Passed	0.00
85	test_pipeline	test_new_customer_total_cannot_exceed_monthly	TC-15	Passed	0.00
86	test_pipeline	test_phone_service_yes_with_no_phone_service_lines	TC-15	Passed	0.00
87	test_pipeline	test_internet_addons_require_internet	TC-15	Passed	0.00
88	test_pipeline	test_internet_customer_cannot_have_no_internet_addons	TC-15	Passed	0.00
89	test_pipeline	test_every_dataset_row_passes_validation	TC-15	Passed	0.42
90	test_pipeline	test_preprocess_input_guards[tenure-abc-Non-numeric]	TC-15	Passed	0.01
91	test_pipeline	test_preprocess_input_guards[tenure-3.5-whole numbers]	TC-15	Passed	0.01
92	test_pipeline	test_preprocess_input_guards[MonthlyCharges-True-Boolean]	TC-15	Passed	0.01
93	test_pipeline	test_preprocess_input_guards[MonthlyCharges-inf-Non-numeric]	TC-15	Passed	0.01
94	test_pipeline	test_preprocess_input_guards[gender-None-Missing value]	TC-15	Passed	0.01
95	test_pipeline	test_preset_is_complete_and_valid[High risk: new fiber customer]	TC-14	Passed	0.00

No.	Module	Test	Test case	Result	Time (s)
96	test_pipeline	test_preset_is_complete_and_valid[Borderline: month-to-month DSL]	TC-14	Passed	0.00
97	test_pipeline	test_preset_is_complete_and_valid[Low risk: loyal family on 2-year contract]	TC-14	Passed	0.00
98	test_pipeline	test_preset_is_complete_and_valid[Low risk: phone-only customer]	TC-14	Passed	0.00
99	test_pipeline	test_preset_prediction_matches_its_label[High risk: new fiber customer]	TC-16	Passed	0.01
100	test_pipeline	test_preset_prediction_matches_its_label[Borderline: month-to-month DSL]	TC-16	Passed	0.02
101	test_pipeline	test_preset_prediction_matches_its_label[Low risk: loyal family on 2-year contract]	TC-16	Passed	0.01
102	test_pipeline	test_preset_prediction_matches_its_label[Low risk: phone-only customer]	TC-16	Passed	0.01
103	test_pipeline	test_default_preset_exists	TC-14	Passed	0.00
104	test_pipeline	test_get_preset_values_returns_a_copy	TC-14	Passed	0.00
105	test_pipeline	test_unknown_preset_raises_key_error	TC-14	Passed	0.00
106	test_pipeline	test_holdout_split_is_the_training_test_split	TC-07	Passed	0.16
107	test_pipeline	test_live_evaluation_reports_all_metrics	TC-06	Passed	0.18
108	test_pipeline	test_live_evaluation_matches_saved_benchmark	TC-17	Passed	0.00
109	test_pipeline	test_confusion_matrix_is_consistent	TC-06	Passed	0.00
110	test_pipeline	test_compute_classification_metrics_on_known_values	TC-06	Passed	0.02
111	test_pipeline	test_metrics_when_model_never_predicts_churn	TC-06	Passed	0.02
112	test_pipeline	test_metrics_input_errors	TC-06	Passed	0.00
113	test_pipeline	test_lower_threshold_catches_more_churners	TC-17	Passed	0.18
114	test_pipeline	test_missing_dataset_file	TC-02	Passed	0.01
115	test_pipeline	test_dataset_missing_columns	TC-02	Passed	0.07
116	test_sprint2	test_score_customer_batch_holdout_sample	TC-18	Passed	0.04
117	test_sprint2	test_score_customer_batch_custom_threshold	TC-18	Passed	0.05
118	test_sprint2	test_explain_customer_risk_attribution	TC-19	Passed	0.00
119	test_sprint2	test_what_if_counterfactual_risk_reduction	TC-19	Passed	0.03
120	test_system_validation	test_tc01_valid_dataset_loads_and_splits	TC-01	Passed	0.05
121	test_system_validation	test_tc02_missing_dataset_raises_clear_error	TC-02	Passed	0.00
122	test_system_validation	test_tc02_dataset_without_target_raises_clear_error	TC-02	Passed	0.02
123	test_system_validation	test_tc02_cli_stops_without_writing_outputs	TC-02	Passed	0.01
124	test_system_validation	test_tc02_invalid_configuration_is_rejected	TC-02	Passed	0.00

No.	Module	Test	Test case	Result	Time (s)
125	test_system_validation	test_tc02_missing_or_broken_params_file_falls_back_to_defaults	TC-02	Passed	0.00
126	test_system_validation	test_tc03_target_separated_and_text_columns_encoded	TC-03	Passed	0.00
127	test_system_validation	test_tc04_xgboost_trains_with_class_weighting	TC-04	Passed	2.98
128	test_system_validation	test_tc05_grid_search_uses_stratified_folds	TC-05	Passed	0.00
129	test_system_validation	test_tc05_every_grid_combination_is_evaluated	TC-05	Passed	1.24
130	test_system_validation	test_tc05_best_hyperparameters_are_reported	TC-05	Passed	17.22
131	test_system_validation	test_tc06_metrics_and_reports_are_written	TC-06	Passed	0.00
132	test_system_validation	test_tc06_every_model_run_is_logged	TC-06	Passed	0.00
133	test_system_validation	test_tc07_test_set_keeps_the_natural_class_ratio	TC-07	Passed	0.02
134	test_system_validation	test_tc07_class_weighting_is_logged_for_each_model	TC-07	Passed	0.00
135	test_system_validation	test_tc08_model_and_encoders_are_saved	TC-08	Passed	0.00
136	test_system_validation	test_tc09_saved_pair_reloads_and_predicts	TC-09	Passed	0.02
137	test_system_validation	test_tc10_requirements_declare_every_third_party_import	TC-10	Passed	0.05
138	test_system_validation	test_tc10_training_run_completes_within_budget	TC-10	Passed	0.00
139	test_system_validation	test_tc11_personal_identifiers_are_not_model_inputs	TC-11	Passed	0.00
140	test_system_validation	test_tc11_secrets_and_environments_are_git_ignored	TC-11	Passed	0.00
141	test_system_validation	test_tc11_no_network_clients_in_the_code	TC-11	Passed	0.04
142	test_system_validation	test_tc11_no_secret_files_are_tracked	TC-11	Passed	0.03
143	test_system_validation	test_tc12_readme_covers_setup_usage_model_and_limitations	TC-12	Passed	0.00
144	test_system_validation	test_tc12_every_module_documents_its_purpose	TC-12	Passed	0.02
145	test_system_validation	test_tc13_split_is_identical_across_runs	TC-13	Passed	0.04
146	test_system_validation	test_tc13_training_is_deterministic	TC-13	Passed	0.68
147	test_system_validation	test_tc23_benchmark_scores_and_logs_every_paradigm	TC-23	Passed	22.26
148	test_system_validation	test_tc23_benchmark_saves_loadable_serving_models	TC-23	Passed	0.03
149	test_system_validation	test_tc23_benchmark_cli_reports_bad_configuration	TC-23	Passed	0.01
150	test_system_validation	test_tc24_single_prediction_is_fast	TC-24	Passed	0.36

No.	Module	Test	Test case	Result	Time (s)
151	test_system_validation	`test_tc25_production_model_matches_training_feature_order`	TC-25	Passed	0.00
152	test_system_validation	`test_tc25_production_payload_is_a_model_dictionary`	TC-25	Passed	0.00

7. Conclusion

152 of 152 automated tests passed and none failed. The raw JUnit XML and the full pytest output are kept in `raw/` as evidence.